Mathematical Modeling Lecture Notes
Chapter 2 – The Modeling Process, Proportionality, and Geometric Similarity

I. Modeling process
   a. Complexities when dealing with the real world
      i. Here is a graph representing the modeling process (p. 60 in text):

      ii. Collect real world data. How do you do this?
         1. Experiment
         2. Survey
         3. Government or other source
      iii. Simplification
         1. Nearly any mathematical description of a real-world problem necessarily entails some simplification.
         2. The real world in all of its complexity is generally too intractable to represent accurately mathematically.
         3. Examples
            - Ignoring air resistance in projectile motion.
            - Constant growth rate factors in population models.
            - Ask students for others.
      iv. Build a model. (We will be discussing lots of different models this term.)
      v. Analysis
         1. How do you do this?
            - Look at the model's outputs.
            - Sensitivity analysis: Change the inputs and see how the outputs change.
         2. Training vs. testing distinction
      vi. Draw mathematical conclusions. What does my analysis mean, mathematically?
      vii. Interpretation. What do the mathematical conclusions mean about the real world?
      viii. Real world explanations or predictions
      ix. Verification
         1. Check to see whether the real-world explanations or predictions make sense.
         2. The verification arrow points back to real-world data because at this point we might need to collect new data based on our current explanations or predictions.
   b. Talk through Problem 2.1 #2 in class: "A retail store intends to construct a new parking lot. How should the lot be illuminated?"
      i. First, come up with a more specific question whose answer addresses the problem at hand but that is more amenable to mathematical analysis.

1. For example, do you want to maximize the amount of light given a budget constraint?
2. Or do you want the light to be uniformly distributed across the lot?
3. Maybe you want to solve a subproblem first, like finding a model that gives light intensity as a function of distance from a source.

ii. What variables affect the behavior in the problem?
iii. Which variables are most important? Which should be neglected?
iv. What might be considered as constants initially?
v. Are there any submodels we should consider in detail?
vi. What data do we need to collect?

c. Have students work through the following problems in groups: "How should a university dining hall be laid out?", "Design an auditorium for a physics lecture," "Where should Puget Sound place a new parking lot?", "How would you create student work schedules in the cafeteria?", "How would you create the math department's course schedule for next semester?" and "Design a new terminal for SeaTac."

II. Proportionality

a. Background
   i. Often when constructing a model you will have a set of data, and you want to find some kind of relationship between the data values.
   ii. A key idea in constructing such a relationship is that of proportionality.
b. The variable y is *proportional* to some function $f(x)$ iff $y = kf(x)$ for some k.
c. Notation: $y \propto f(x)$
d. This implies that a graph of y vs. $f(x)$ will look like a straight line through the origin having slope k.
e. Work Problem 2.2 #10 (modified) to see whether the data given support the proportionality model $y \propto x^3$.
   i. The graph doesn't look very linear.
   ii. It actually looks more like a power graph.
   iii. In fact, it's more likely that $y \propto x^2$. (Test this.)
f. Caution: In order for proportionality to hold, the line of y vs. $f(x)$ really does need to go through the origin.

III. Geometric similarity

a. Two objects are *geometrically similar* if there is a one-to-one correspondence between points of the objects such that the ratio of the distances between corresponding points is the same for all possible pairs of points.
b. A *characteristic dimension* is a dimension that defines the scale of the system. It can be used to express many other properties of the model.
c. Suppose we have two geometrically similar cylinders with the first being scaled by a factor of k to produce the second.
   i. Let the radius be a *characteristic dimension* for the cylinders; i.e., one in which we can express other properties of the cylinder.
   ii. Let r_1 be the radius of one cylinder and r_2 the radius of the other so that $k = r_2/r_1$ is the ratio of the radii.
   iii. Similarly, $h_2/h_1 = k$ because of the definition of geometric similarity.
   iv. Are there other measures of a cylinder whose ratios would give the same constant k for geometrically similar cylinders?
      1. Diameter, as $d_2/d_1 = 2r_2/(2r_1) = k$.

2. Circumference, as $c_2/c_1 = 2\pi r_2/(2\pi r_1) = k$.

v. Are there measures of a cylinder whose ratios do not give k for geometrically similar cylinders?

1. Surface area
   - We have $S_1 = \pi r_1 h_1$ and $S_2 = \pi r_2 h_2$.
   - Thus $S_2/S_1 = r_2 h_2/r_1 h_1 = k^2$.
2. Volume: $\dfrac{V_2}{V_1} = \dfrac{\pi r_2^2 h_2}{\pi r_1^2 h_1} = k^3$.

d. More on surface area and volume

i. Under the geometric similarity assumption surface area is always proportional to the square of the radius.

ii. For example

1. With the cylinders we have $S_2/S_1 = k^2 = r_2^2/r_1^2$.
2. Thus $S_2/r_2^2 = S_1/r_1^2$.
3. The only way these expressions can be equal for all geometrically similar cylinders is for their ratios to be constant.
4. Thus $S \propto r^2$ for all such geometrically similar cylinders.
5. Surface area is also proportional to the square of the height, or the diameter, or the circumference, under the geometric similarity assumption.

iii. And, not surprisingly, volume is proportional to the cube of the radius or height, etc.

iv. These imply that, under the geometric similarity assumption, if we're interested in some property y such that $y = f(r, S, V)$ then we can actually express y solely as a function of r through $y = g(r, r^2, r^3)$.

e. Project 2.3 #1: In the TV show *Superstars* the top athletes from various sports compete against one another in a variety of events. The athletes vary considerably in height and weight.

i. To compensate for this in the weight-lifting competition, the body weight of the athlete is subtracted from his lift. What kind of relationship does this suggest?
   - It suggests that $L - W = k$, where L is the max lift, W is the weight, and k is some constant.

ii. Test this assumption using the following winning lifts data from the 2000 Olympics:

Men		Women	
Max weight (kg)	Lift (kg)	Max weight (kg)	Lift (kg)
56	305	< 48	185
62	325	$48 - 53$	225
69	357.50	$58 - 63$	242.50
77	376.50	75	245
85	390	> 75	300
94	405		
105	425		
> 105	472.5		

1. Are the missing weight classes for women errors or missing data?
2. Graphing the data on men and women indicates that they should be analyzed separately.

3. A graph of $L - W$ (for men) does not appear to give a constant value at all. The higher weight classes are advantaged.

iii. Physiological arguments have been proposed that suggest that the strength of a muscle is proportional to its cross-sectional area. Using this submodel for strength, construct a model relating lifting ability and body weight.
   1. The physiological argument gives $L \propto A$.
   2. We also have $W \propto V \propto l^3$ for some characteristic dimension l.
   3. Then, as $A \propto l^2$, we now have $L \propto W^{2/3}$.

iv. What are the assumptions we just made?
   1. We assume geometric similarity among lifters in different classes.
   2. We assume that the cross-sectional area of a lifter's muscles is proportional to the lifter's cross-sectional area in general.

v. Test the model just obtained.
   1. We have $L = kW^{2/3}$.
   2. A graph of L vs. $W^{2/3}$ does give an approximately linear relationship.

vi. Use the model to obtain an alternative handicapping equation than $L' = L - W$.
   1. The model claims that if we divide L by $W^{2/3}$ we should obtain a constant.
   2. Thus an appropriate handicapping equation is $L' = L/W^{2/3}$.
   3. Under this model the winner is the person in weight class 69.

vii. Critique the model
   1. People in the middle weight classes might be advantaged under this model.
   2. The winning lifts from a competition may not be the best choices for data; perhaps the average weight lifted by someone in each weight category would be better.
   3. Maybe we should have used the midpoint of each weight class rather than the maximum.
   4. We're doing training and testing on the same data set.
   5. The data set is really small.

Homework:

- Section 2.1: #7, 9.
- Section 2.2: #8, 12.
- Section 2.3: #2, 4.